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Q1. Harshad number

ANSWER

```
n=int(input(""))
count =0
for i in range(1,num+1):
if num%i==0:
count=count+1
print(count)
```

Q2. GCD

CODE

```
a=int(input())
b=int(input())
while b!=0:
    temp=b
    b=a%b
    a=temp
print(a)
```

INPUT

```
12
18
```

OUTPUT

```
6
```

Q3. sum to a target value

ANSWER

```
set()
1st:
-num
in seen:
seen.
pairs
find_pairs
```

Q4. Bitwise Operators

ANSWER

```
data type
a->int
b->int
c->int
result->int
step by step
a=0b1101=13
b=0o25=21
c=0*1a=26
a&b=13&21=5
result=5|26=31
output
a=12,b=21,c=26,result=31
ob11111 0o37 0x1f
7 62-32
```

Q5. binary search**CODE**

```
def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low ≤ high:
        mid=(low+high)//2
        if arr[mid]== target:
            return mid
        if arr[low]≤ arr[mid]:
            if target≥arr[low]and target<arr[mid]:
                high=mid-1
            else:
                low=mid-1
        else:
            if target > arr[mid]and target≤arr[high]:
                low=mid+1
            else:
                high=mid-1
    return-1
data=[4,5,6,7,0,1,2]
t=int(input("target:"))
result=rotated_binary_search(data,t)
print(result)
```

OUTPUT

```
Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192521249RC_6a62471bd3bbb4.34057364\main.py", line 21, in <module>
    result=rotated_binary_search(data,t)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^
NameError: name 'rotated_binary_search' is not defined. Did you mean: 'roated_binary_search'?
```